

Human First: Introduction to AI and Human-AI Teaming

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Why this Approach?

Forecasting involves more than models

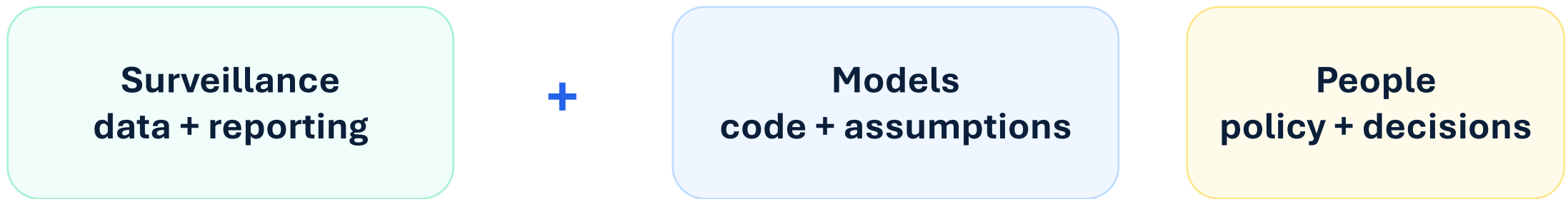
A forecast is not simply a model output. It is the product of a public-health question, surveillance data, software, assumptions, scientific judgment, and an eventual decision.

- AI can help clarify the forecasting question and surface missing context.
- It can audit data and draft reproducible cleaning code.
- It can help build, compare, and document candidate models.
- Humans must validate assumptions, intermediate outputs, and uncertainty.
- The final scientific claim and decision remain human responsibilities.

The forecasting system—not the AI alone—is what must perform.

Forecasting is a sociotechnical system

A forecast is created inside a larger network of technology, expertise, institutions, and decisions.



- Reporting systems shape the observed data.
- Analysts shape the model and interpretation.
- Decision context determines whether a forecast is useful.

Teaming provides a construct and mechanism to integrate data, models, and human decision-making.

A Brief Primer on Artificial Intelligence

AI is a family of capabilities

Artificial intelligence refers to computational systems that perform functions associated with intelligent behavior. It is a broad field containing different methods, capabilities, and levels of autonomy—not one technology.

- Classification and prediction: estimate a class, count, or probability.
- Recommendation and optimization: compare alternatives against a goal.
- Generation: produce language, code, images, or other content.
- Planning and action: carry out a bounded sequence of steps using tools.

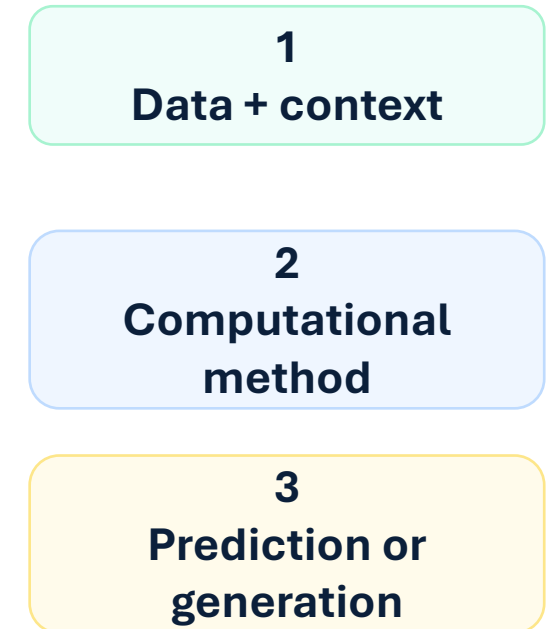
AI turns inputs into outputs

Most AI systems transform available information into an output through learned or specified relationships.

Examples:

- Surveillance data → risk estimate
- Historical counts → forecast
- Text prompt → generated explanation

A simplified view



Machine learning and GenAI differ

Both use learned patterns, but they support different kinds of work.

Machine learning

- Estimates relationships in structured data
- Classifies or predicts outcomes
- Often produces a number, class, or probability

Generative AI

- Produces language, code, images, or other content
- Generates plausible continuations
- Often works through natural-language interaction

AI is not human intelligence

Sophisticated output can look intentional and informed, but fluent language does not imply human experience, causal understanding, or public-health judgment.

- AI does not automatically understand meaning, mechanism, or causality.
- It does not experience the consequences of a public-health decision.
- It can reproduce learned patterns without grounding them in the current setting.
- It does not assume accountability simply because it produced the output.

AI has developed in waves

The field repeatedly advances through new representations, data, and computing power.

1950s–1980s
**Symbolic AI + expert
systems**

1990s–2010s
**Statistical ML + deep
learning**

2020s
**Foundation models
+ agents**

- Periods of rapid progress have been followed by AI winters.
- The enduring tension is capability versus expectation.

AI history is also expectation history

Every wave produces genuine advances—and claims that exceed the evidence.

What improves

- Computational capability
- Range of tasks
- Accessibility
- Speed and scale

What persists

- Boundary failures
- Context sensitivity
- Human overexpectation
- Need for verification

Several forces converged

The current expansion of AI did not come from one breakthrough. Several technical and social forces converged to make advanced capabilities powerful, reusable, and easy to access.

- Large-scale digital data supported broader pattern learning.
- Greater computing capacity made larger models feasible.
- Improved architectures expanded the range of learnable relationships.
- Foundation models could be reused across many tasks.
- Natural-language interfaces placed those capabilities in everyday workflows.

AI is not one tool

Different tasks call for different capabilities and different levels of oversight.

Predictive

- Estimate future disease outcomes
- Produce probabilities or counts
- Evaluate with held-out data

Generative

- Explain and critique
- Draft code or documentation
- Explore alternatives

Agentic

- Work across files
- Run tools and tests
- Requires stronger boundaries

AI can miss the mark confidently

AI can produce a polished, complete-looking answer before the underlying analysis is valid. Confidence and fluency make these failures easy to miss during fast analytical work.

- Invented facts, citations, variables, or data sources.
- Incorrect code that still runs and produces plausible output.
- Data leakage or inappropriate train–test separation.
- Unsupported causal interpretations of observational patterns.
- Excessive certainty and weak communication of uncertainty.

Fluency is a presentation property—not evidence of correctness.

Human-AI Teaming

Not every AI use is teaming

Human–AI relationships differ in agency, interdependence, and coordination.

Tool

- Executes a defined operation
- Limited initiative
- Human directs the function

Assistant / collaborator

- Responds and iterates
- Supports a shared product
- Moderate coordination

Teammate

- Distinct role
- Some agency
- Interdependent contribution

Human–AI teaming has boundaries

A human–AI team contains humans and autonomous agents working interdependently toward a shared objective. The relationship is defined by observable work—not by conversational language or a product label.

- One or more humans and at least one AI agent.
- A shared goal that organizes collective activity.
- Distinct roles and unique contributions.
- Interdependence in activity, information, or outcomes.
- Performance evaluated at the team level.

O'Neill, McNeese, Barron, & Schelble (2022)

Shared goal + distinct roles + agency + interdependence distinguish teaming from basic automation.

Is this actually a team?

Use observable properties rather than metaphors or marketing language.

Ask whether the AI:

- Contributes something unique
- Has decision latitude
- Changes what the human does

Three essential tests

1
Shared goal

2
Distinct roles

3
Interdependent work

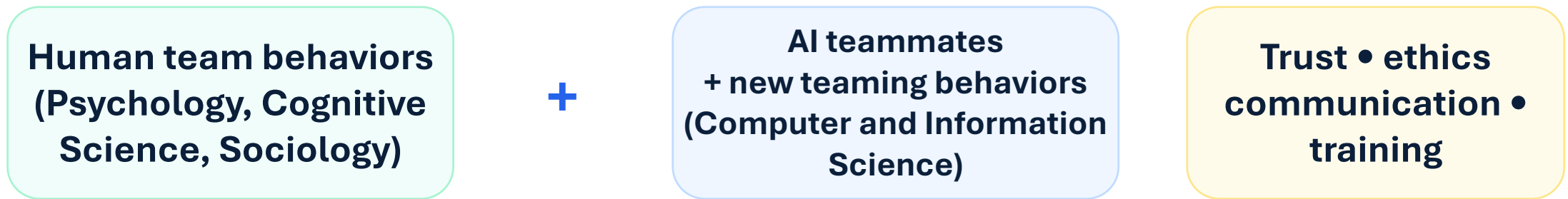
The team is the unit of analysis

AI performance tells us what a system can do in isolation. Human–AI teaming research asks whether the combined system coordinates effectively, maintains human understanding, and improves the work.

- Roles determine who contributes, checks, and remains accountable.
- Communication determines whether the right information arrives at the right time.
- Trust determines when people rely, verify, or intervene.
- Training changes expectations and coordination behavior.
- Context determines whether any of those processes are effective.

Human–AI teaming is a cumulative science

Moves from foundational teamwork science to the design of AI teammates.



- Computational buildout of AI teammates or models oriented for collaboration
- Controlled experiments and simulations
- Contextual inquiry and participatory design
- Healthcare, emergency response, aviation, defense, education, and collaborative work

Teams perform two kinds of work

An AI can be technically capable at taskwork while being poor at teamwork.

Taskwork

- Clean data
- Write code
- Fit models
- Calculate metrics
- Produce forecasts

Teamwork

- Share information
- Coordinate roles
- Monitor progress
- Anticipate needs
- Recover from errors

Coordination depends on shared knowledge

Team cognition is the knowledge that enables members to understand the task, anticipate one another, coordinate roles, and recognize when the current plan needs to change.

- Goal: What decision will this forecast support?
- Task: What target, horizon, data, and model are in scope?
- Roles: What will the human do, and what may the AI do?
- Sequence: What happens next, and what evidence is required?
- Intervention: When should the human stop, correct, or redirect the AI?

Context must be made explicit

Human teammates rely on experience, conventions, and implicit cues. AI systems may not possess that context unless we deliberately represent it in prompts, files, examples, and project documentation.

- Define variables, units, and missing-value conventions.
- Document reporting delays, revisions, and known anomalies.
- State analytical constraints and quality criteria.
- Preserve decisions in prompts, code comments, and project files.
- Update the shared context whenever the task or assumption changes.

HAT research evaluates more than accuracy

A major review identified 76 empirical studies that met explicit human–AI teaming criteria. The unit of evaluation was the combined team—not only the algorithm.

- Team and individual performance.
- Workload and situation awareness.
- Trust, reliance, and human perceptions.
- Coordination and communication quality.
- Shared mental models and team cognition.
- Adaptation, error recovery, and training effects.

O'Neill et al. (2022)

The science asks whether the team works—not only whether the algorithm works.

HATs are not automatically better

Early HATs often underperformed human-only teams because agents were built for tasks rather than teamwork.

Where HATs struggle

- Weak anticipation
- Reactive behavior
- Poor role fit
- Limited communication

What determines results

- Task complexity
- Agent reliability
- Team composition
- Training and design

Trust should track evidence

The goal is calibrated reliance—not maximum trust.

Overtrust

- Accept code without running it
- Ignore contradictory evidence
- Confuse confidence with correctness

Undertrust

- Reject useful assistance
- Repeat all work manually
- Miss valid AI capabilities

Reliance should change with task, evidence, and demonstrated capability.

Do not build an ideal human substitute

People often expect an AI teammate to combine perfect machine performance with the flexibility, empathy, and contextual understanding of an ideal human teammate. That expectation is neither realistic nor useful for design.

- Design AI teammates around distinct, demonstrated capabilities.
- Use human teamwork science as a foundation—not a complete blueprint.
- Make consequential differences visible so people can coordinate around them.

A forecasting team divides responsibility

Responsibility shifts across the pipeline; it is not divided equally.

Human leads

- Scientific question
- Epidemiological context
- Standards and final claims

AI supports

- Audits and code drafts
- Pattern exploration
- Diagnostics and documentation

Together check

- Assumptions
- Intermediate outputs
- Evidence for refinement

Teaming happens through handoffs

A practical HAT is an evidence-based sequence of action, inspection, and adjustment.

Example:

- Human defines 1–4 week target
- AI flags duplicate records
- Human resolves reporting rule
- AI implements a bounded change

One interaction cycle

1
Frame + clarify

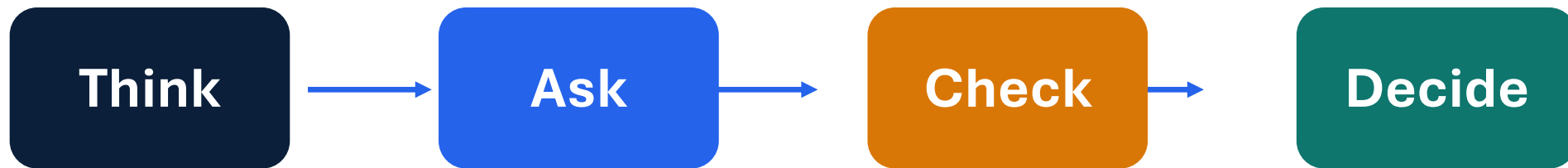
2
Implement + run

3
Inspect + revise

Shared goal + distinct roles + monitoring + adjustment = teaming in practice.

Human judgment organizes the system

AI capability becomes valuable when humans frame, coordinate, evaluate, and remain accountable.



Human judgment + AI capability + disciplined evaluation = better forecasting